

**INFLUENCE OF SCHOOL INFRASTRUCTURES ON THE COGNITIVE, AFFECTIVE
AND PSYCHOMOTOR ABILITIES OF STUDENTS IN PUBLIC
SECONDARY SCHOOLS IN ENUGU STATE,
NIGERIA**

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Abstract

The purpose of this study was to determine the influence of school infrastructure on the cognitive, affective and psychomotor abilities of students in public secondary schools in Enugu State. Three research questions guided the study and three null hypotheses were tested at .05 level of significance. Descriptive survey research design was used for this study. The population for the study comprised all the 112,481 secondary school students in the 298 public secondary schools in Enugu State. The researcher used proportionate stratified random sampling technique to draw 5% of male and female secondary school students from the 298 public secondary schools in Enugu State. First, the researcher got the total population of 37,493 male and 78,988 female students. From this the researcher drew 5% of male (175) and female (222) secondary school students, from each stratum respectively. The derived sample was summed up to get the sample size of 397. A structured questionnaire was used for data collection. The instrument was validated by three experts from the Faculty of Education, Enugu State University of Science and Technology. Cronbach Alpha reliability estimate was used to determine the internal consistency of the instrument. The overall reliability index stood at .77 indicating that the instrument was highly reliable and suitable for the study. The researcher administered the instrument directly to the 397 respondents using five research assistants. Out of the 397 administered copies of the instrument only 318 copies were correctly filled and retrieved. 79 copies were not correctly filled. Mean and standard deviation were employed in answering the research questions that guided the study, while t-test at 0.05 level of significance was used to test the null hypotheses formulated for the study. It was found that school infrastructures influence the cognitive ability, and psychomotor ability of secondary school students in Enugu State to a high extent. The finding revealed that school infrastructures influence the affective ability of secondary school students in Enugu State to a low extent. This is supported by the fact that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructure influence the cognitive ability, affective ability and psychomotor ability of secondary school students in Enugu State. Based on the findings, the researcher recommended among others that deliberate efforts should be made by Ministry of Education at supervising school infrastructures in secondary school in Enugu State.

Keywords: School Infrastructures, Cognitive Abilities, Affective Abilities, Psychomotor Abilities, Students, Secondary Schools

Introduction

Secondary education is one aspect of educational institutions in Nigeria that is designed specifically to train and prepare students for middle-level services in both manufacturing and service industries. Secondary education does not only occupy an important place in the Nigeria Education System but also serves as a link between the primary and tertiary levels of education; and because of its central position, its programmes have functional roles such as giving students access to higher education as well as preparing them for work. Ede (2021), stated that secondary education provides avenue for interaction among human being with divergent needs, interest and motives. Enose (2020), summed it up by noting that educational organization such as secondary school system exists in a symbolic relationship with the environment while utilizing both human and material resources for the production of educated, socialized graduates. Significant to this process is the school infrastructure.

School infrastructures refer to the indirect teaching support facilities such as offices, cafeteria, acoustics, toilets, laundry, mowers, residential halls, common rooms, cleaning materials ground and similar items that satisfy the individual's physical and emotional needs. They are used to increase instructional effectiveness, improve the cleanness, orderliness and safety of facilities, reduce the operational cost and life cycle cost of a building, extending the useful life of a building, increasing efficiency and effectiveness of the staff and students, improve building appearance, use data collection and analysis for decision making

(Asiabaka, 2023). According to Brookshire (2021), school infrastructures are the essential physical and material resources and facilities, including buildings, classrooms, libraries, laboratories, playgrounds, furniture, and technology, that directly and indirectly support the teaching and learning process. This implies that school infrastructures seem to constitute the major components of both direct and indirect action elements in the environment of learning.

School infrastructures encompass both the fundamental structures like buildings and the essential services and amenities, such as water, electricity, and furniture, that contribute to a safe, effective, and conducive environment for education. Adeogun and Olisaemeka (2017) in another dimension noted that the key elements of school infrastructures include well-equipped classrooms, functional science labs, well-stocked libraries, effective sanitation, adequate lighting, and sufficient ventilation, all of which contribute to a higher quality of education and can lead to better student attendance, concentration, and overall performance in school. The availability of school infrastructures such as classes, desks, toilets and good learning facilities in different countries worldwide have been prioritized for the intention of enhancing quality education (Nurabadi, 2020). Unfortunately, in Nigeria, many schools, especially in Enugu State, suffer from poor infrastructures, contributing to the widening gap in academic performance among schools in Enugu State (Ihebom & Uko, 2020). Atkin (2019) revealed that the status of school infrastructures in Enugu

State varies, with studies indicating that public secondary schools suffer from dilapidated buildings, insufficient classrooms, and inadequate facilities like toilets and clean water among others. Ekundayo (2020) pointed out that in most of the nation's secondary schools, teaching and learning take place under a most unconducive environment, lacking the basic materials and thus hindered the fulfillment of educational objectives. In the words of Azubuike, (2024), the level of dilapidation in the schools is monumental.

It is established that libraries lack current and well-balanced stock that is capable of meeting the information needs of teachers and students, teaching and laboratory materials are also in short supply. In the view of Olawale (2024), the facilities in educational institutions in Enugu State are too poor to be compared with other societies of the world. Moja (2024) stated that the basic infrastructures in schools in Enugu State such as classrooms, laboratories, workshops, sporting facilities, equipment, libraries were in a state of decay. The author added that physical condition of most schools was reported to be pathetic. There is an indication that vast majority of Nigeria public schools especially secondary schools are in terrible shape; buildings are dilapidated, and equipment is either obsolete or unavailable. Gbinde (2023) revealed that instructional materials are lacking, students construct their chairs and lockers and bring them to school else, they sit with their mates who may either be willing or unwilling to share their seats, sometimes they sit on bare floor. In line with the above, Usombo (2023) stated that public secondary schools have

suffered untold neglect and the resultant effect has been that of low quality school leavers. Atkin (2019) stressed that it is important for the schools to have well improved physical infrastructures since having such infrastructures within and outside the classrooms will make both teachers and students comfortable in the teaching and learning processes.

The presence of good infrastructures also motivates students to stay in schools as the school environment becomes welcoming and attractive. Jideowolabi (2024) suggested that infrastructural materials such as school buildings, furniture and fittings, libraries, audiovisuals and laboratories need to be in place, and a regular allocation of resources, including textual, stationeries, chemicals and chalkboards tools are essential for enhancing students' abilities in schools. Ugwu (2019), averred that students' abilities refers to how well a student is accomplishing his tasks and studies with reference to his cognitive, affective and psychomotor.

Cognitive ability refers to how an individual processes information, recognizes patterns and analyzes problems. According to Iloka (2024), cognitive ability focuses on six intellectual skills which include; knowledge, comprehension, application, analysis, synthesis and evaluation. Ekanem (2021) saw cognitive domain as a collection of natural skills that include such things as listening, perception and attention to detail. Ugwu (2019), noted that cognitive domain is concerned with the intellectual side of learning. It refers to knowledge attainment and mental/intellectual processes. They include skills such as reasoning, memory and concentration. Onovo (2022), asserted

that the stronger the cognitive ability are established in an individual, the easier it is for the individual to learn. On the contrary, Udoka (2021) posited that negative experiences within a students' environment such as school environment has significant impact on the cognitive ability of such student. There is therefore need to ascertain the extent to which school infrastructure influences the cognitive ability of secondary school students in Enugu State, as the gap created by this is long overdue. Next to cognitive ability is the affective ability.

Affective ability of learning represents skills that foster appropriate emotional responses. It characterizes the emotional arena reflected by learners' beliefs, values and interest. The affective ability includes the manner in which we deal with things emotionally, such as feelings, values, appreciation, enthusiasms, motivations and attitudes (Ugwu, 2019). Ubaka (2018), noted that like the cognitive domain, the areas of emotional response in the affective domain are; receiving, responding, valuing, organizing and characterizing. Being that the affective domain has to do with emotions and feelings, Lehrer (2018) noted that school infrastructures have serious impact on the human emotion particularly the students. Abugieye (2020) revealed that school infrastructures enhance students interest will assist them with better ways of handling things emotionally especially as it affects their academics. Study by Yangambi (2023) revealed that availability of school physical facilities and the quality of the school learning environment was said to be powerful factors influencing student success

in the emotional and psychomotor domains. Edoka (2022) reported that educational resources which are physical, human, material and symbolic resources influence the academic behaviour of students. The author, added that students are attracted by physical structures which often motivate their interest in learning. Okoroafor (2017), stated that poor school infrastructure not only influence the adjustment and potential of students but lead to drop-out from school as a result of interest and experiences. This call for a serious concern, however, the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State, is a gap of serious concern. Close to affective ability is the psychomotor ability.

Psychomotor ability reflects learning behaviour achieved through neuromuscular motor activities. It focuses on physical abilities. Udoka (2021), posited that psychomotor ability helps people perform physical tasks in daily life and at work. The areas of the psychomotor domain include; perception, set, guided responses, mechanism, complex overt response, adaptation and origination. Ugwu (2019), stated that the psychomotor ability includes physical movement, coordination, use of the motor-skills areas. Ugwu added that the development of these abilities requires practice and it is measured in terms of speed, precision, distance, procedures, or techniques in execution. Chiege (2020) stated that secondary school students' ability to acquire abilities like psychomotor development has been shown to be strongly impacted by the quality of the physical environment in schools. Nkannem (2021)

noted that without these resources, the classroom environment will not be conducive for effective teaching and learning, which will cause students to become uncomfortable and divert their attention from their studies. Study by Iyere (2023) showed that availability of suitable materials and facilities for psychomotor activities is crucial for the development of motor skills in students. There is therefore need to explore the extent to which school infrastructures influence the cognitive, affective and psychomotor abilities students in secondary schools in Enugu State as the gap created by this is long overdue, especially with regards to gender of public secondary school students in Enugu State.

Gender is a socially learned behaviour and expectation associated with male and female. It is described as the biological sex of individuals in terms of being male or female. In Nigerian society, there are differences and inequalities between women and men in responsibilities assigned, activities taken, access to and control over resources as well as possession of qualities (Abayi, 2022). Male and female significantly differ in their cognitive abilities (Ugwu, 2022). Ede (2018), noted that male students develop better psychomotor ability more than their female counterparts. Okoro (2021), posited that male and female students do not differ significantly in their perception and educational abilities. Ukaïke (2023), argued that gender does not influence students' academic performance in secondary schools. This may be probably due to the assumption that the male and female genders possess equal right. The issue of gender has gained much attention

with little or no conclusion especially as regards the school infrastructure and the cognitive, affective and psychomotor abilities of students in secondary schools in Enugu State.

This researcher is concerned that if adequate attention is not paid to ascertain the influence of school infrastructure on the cognitive, affective and psychomotor abilities of secondary school students in Enugu State, education delivery might be jeopardized. This is because no meaningful teaching and learning can occur in an environment characterized by poor school infrastructure. In this case, both the teachers' productivity and students' academic performance will continue to be adversely affected. It is equally a source of worry that literatures available to the researcher did not provide any empirical data on the extent to which school infrastructure influence the cognitive, affective and psychomotor abilities of secondary school students in Enugu State. It is therefore against this backdrop that the researcher is motivated to carry out this research on the influence of school infrastructure influence on the cognitive, affective and psychomotor abilities students in Enugu State. This constitutes the gap to be filled in this study.

Statement of the Problem

It has been observed that secondary schools in Enugu State are faced with poor and inadequate school infrastructures, despite governments interest and yearnings for experiential learning at all levels of Education in Enugu State. It is a thing of serious concern that the essential physical and material resources and facilities, including buildings, classrooms, libraries,

laboratories, playgrounds, furniture, and technology, that directly and indirectly support the teaching and learning process are not adequately provided for. Consequently, most public secondary schools in the state, have dilapidated buildings, insufficient classrooms, and inadequate facilities like laboratories, libraries, poorly ventilated classrooms, toilets and clean water. These deficiencies seem to be linked to poor student outcomes, particularly in secondary schools, where infrastructure investment is severely lacking. It is acknowledged that government had tried to improve facilities at some public secondary schools in Enugu State. A good number of schools however still lack requisite learning facilities, while some cannot boast of electricity supplies to them, many do not have toilet facilities and many which have, their electrical fittings and window glasses have been removed by hoodlums. Consequently, students in the Enugu State are faced with unattractive school buildings, crowded classrooms, non-availability of playing ground nor recreational facilities and surroundings that have no aesthetic beauty, not to talk about water and toilet facilities which is alleged to have effects on their performances in class. The researcher is concerned that no empirical evidence known to the researcher has ever revealed the influence of school infrastructures on the cognitive, affective and psychomotor abilities of secondary school students in Enugu State. This gap in knowledge informed the researcher's choice on this topic of study.

Purpose of the Study

The main purpose of this study was to determine the influence of school infrastructures on the cognitive, affective and psychomotor abilities of students in public secondary schools in Enugu State. Specifically, this study sought to;

1. ascertain the extent to which school infrastructures influence the cognitive ability of secondary school students in Enugu State.
2. find out the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State.
3. determine the extent to which school infrastructures influence the Psychomotor ability of secondary school students in Enugu State.

Research Questions

The following research questions guided the study.

1. To what extent do school infrastructures influence the cognitive ability of secondary school students in Enugu State?
2. To what extent do school infrastructures influence the affective ability of secondary school students in Enugu State?
3. To what extent do school infrastructures influence the psychomotor ability of secondary school students in Enugu State?

Hypotheses

The following null hypotheses were tested at .05 level of significance

HO₁: There is no significant difference in the mean ratings of male and female students on the extent to which

school infrastructures influence the cognitive ability of secondary school students in Enugu State.

HO₂: There is no significant difference in the mean ratings of male and female students on the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State.

HO₃: There is no significant difference in the mean ratings of male and female students on the extent to which school infrastructures influence the psychomotor ability of secondary school students in Enugu State.

Method

Descriptive survey research design was used for this study. Descriptive survey research design, according to Nworgu (2015), is one in which a group of people or items are studied by collecting and analyzing data from only a few people or items considered to be representative of the entire group. A descriptive survey research is concerned with specified population of persons, items or situations, in a defined geographical location. It involves the collection of relevant data for or about the population that enables the description of the person, items or situations the way they are (Kothari, 2021). The descriptive survey research design was considered suitable since the study solicited for information from the respondents directly and afforded all the respondents equal chance of being chosen for the study. The population for the study comprised all the 112,481 secondary school students in the 298 public secondary schools in Enugu State. The population was made up of 37,493 male and 74,988 female

students in the public secondary schools in Enugu State. The sample size for this study is 397 public secondary school students in Enugu State. The sample was made up of 175 male and 222 female secondary school students in Enugu State. The researcher used proportionate stratified random sampling technique to draw 5% of male and female secondary school students from the 298 public secondary schools in Enugu State. First, the researcher got the total population of 37,493 male and 78,988 female students from the data collected from the PPSMB. From this the researcher drew 5% of male (175) and female (222) secondary school students, from each stratum respectively. The derived sample was summed up to get the sample size of 397. This was in line with Uzoagulu (2011), who stated that when the respondents' population is in thousands, 5% of the entire population can serve as the sample.

A structured questionnaire titled "Influence of School Infrastructures on Students' Ability Questionnaire (ISISAQ), developed by the researcher was used for data collection. The instrument had two sections; A and B. Section A contains the respondents bio- data while section B is divided into four clusters with 37 items, structured to assist the researcher in providing answers to the research questions that guided the study. Cluster 1 is on the cognitive ability with 11 items, Cluster 2 is on affective ability with 15 items, while cluster 3 is on psychomotor ability with 11 items. The response format for clusters was a 4 point scale type with Very High Extent (VHE), High Extent (HE), Low Extent (LE) and Very Low Extent (VLE). In order to

ensure the validity of the instrument, draft copies of the instrument together with the research topic, purpose of the study, research questions, hypotheses and the developed instrument were given to three experts. Two experts from the Department of Educational Management, one experts from the Department of Mathematics and Computer Education, Measurement and Evaluation Option, all from Faculty of Education, Enugu State University of Science and Technology, Enugu. The experts were requested to assess the relevance, adequacy, suitability and comprehensiveness of the items in addressing the research questions as well as the clarity of the instruction to the respondents. The initial 23 generated items were increased to 37 items as suggested by the validators, while double barreled questions and grammatical errors were corrected as well. The validators' comments were used to draft the final instrument that was used for data collection. The researcher conducted a trial test using 20 respondents from secondary schools in Ebonyi State. The respondents were assured of complete confidentiality of all information they supplied. They were allowed to complete the instrument at their own convenience. The responses to the various items of the questionnaire were used in computing its reliability using Cronbach Alpha reliability estimate. The choice for Cronbach Alpha is in line with the assertions of Daniel (2019) that Cronbach Alpha is applied when multiple likert-scale questions are used in a questionnaire. Thus, the researcher used Cronbach Alpha reliability estimate to determine the internal consistency of the

instrument. Reliability indices of .75, .74, and .89 were obtained for clusters 1, 2, and, 3 respectively. The overall reliability index stood at .79 indicating that the instrument was highly reliable and suitable for the study.

The researcher administered the instrument directly to the 397 respondents using five research assistants. Each research assistant covered a particular education zone and the researcher as well. The research assistants were briefed by the researcher on the conduct of the exercise, during which they were acquainted with the purpose of the study and how the questionnaire should be administered and collected. Instruments were administered and retrieved immediately from the respondents. However, out of the 397 administered copies of the instrument only 318 copies were correctly filled and retrieved. 79 copies were not correctly filled. Mean and standard deviation were employed in answering the research questions that guided the study, while t-test at 0.05 level of significance was used to test the null hypotheses formulated for the study. On the decision rule, real limits of the numbers were applied. Thus, the upper and lower limits of the mean were used to interpret the respondents mean scores as follows:

Very High Extent (VHE) = 3.50 — 4.00

High Extent (HE) = 2.50 — 3.49

Low Extent (LE) = 1.50 — 2.49

Very Low Extent (VLE) = 1.00 — 1.49

The null hypotheses was not rejected when the significant level is less than .05 and was rejected when the significant level is equal or more than .05 level of significance.

Results

Research Question 1

To what extent do school infrastructures influence the cognitive ability of secondary school students in Enugu State?

Table 1: Mean Ratings and Standard Deviation of Male and Female Students on the Extent to which School Infrastructures Influence the Cognitive Ability of Secondary School Students in Enugu State.

S/N	Influence of school infrastructures on the cognitive ability of secondary school students include;	Male N=155		Female N=163		Overall		Decision
		$\bar{X}_1$	SD ₁	$\bar{X}_2$	SD ₂	$\bar{X}_G$	SD _G	
1	enhancing students cognitive ability	3.08	0.62	3.07	0.61	3.08	0.62	HE
2	motivating students to reason out important points from materials read	3.38	0.49	3.39	0.49	3.39	0.49	HE
3	motivating students to comprehend information in the materials read	3.23	0.58	3.23	0.58	3.23	0.58	HE
4	motivating students to obtain higher grade in school	2.93	0.73	2.93	0.73	2.93	0.73	HE
5	enhancing students' to develop problem solving skills	3.31	0.61	3.30	0.61	3.31	0.61	HE
6	strengthening students ability to memorize concepts taught	3.39	0.74	3.37	0.75	3.38	0.74	HE
7	enhancing students' ability to recall concepts taught	3.46	0.50	3.45	0.50	3.46	0.50	HE
8	enhance students ability to recall concepts read	3.54	0.50	3.53	0.50	3.53	0.49	VHE
9	reinforcing students ability to discuss lesson learnt from materials studied	3.32	0.60	3.31	0.61	3.31	0.61	HE
10	enhancing students ability to illustrate a given concept	3.00	0.79	3.00	0.79	3.00	0.79	HE
11	enhancing students' ability to formulate new ideas	3.15	0.54	3.15	0.54	3.15	0.54	HE
Cluster Mean		3.25		3.25		3.25		HE

Data presented in Table 1 shows that all the respondents' responses indicated high extent with their mean points above the cut-off point in the real limit of numbers. The Standard deviation values were small indicating that the respondents' responses are closely clustered around the mean, signifying that their responses are homogenous. The overall cluster mean of 3.25 indicated that the items are the influence of school infrastructure on the cognitive ability of secondary school students in Enugu State. This implies that school infrastructures influence the cognitive ability of secondary school students in Enugu State to a high extent.

H0₁ There is no significant difference in the mean ratings of male and female students on the extent to which school infrastructures influence the cognitive ability of secondary school students in Enugu State.

Table 2: Summary of t-test Analysis of Mean Response Scores of Male and Female Students on the Extent to which School Infrastructures Influence the Cognitive Ability of Secondary School Students in Enugu State.

VARIABLES	N	t	df	Sig. (2tailed)	Mean Difference	Std. Error Difference	Decision
Male Student	155	.123	316	.90	.03831	.31135	Do not reject H0 ₁
Female Student	163						

NS= Not Significant

The result of t-test analysis in Table 2 shows that the t-value at .05 level of significant and 316 degree of freedom for the items is .12 with a significant value of .90. Since the significant value of .90 is more than the .05 level of significant the null hypothesis is not significant. This means that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructure influence the cognitive ability of secondary school students in Enugu State.

Research Question 2

To what extent do school infrastructures influence the affective ability of secondary school students in Enugu State?

Table 3: Mean Ratings and Standard Deviation of Male and Female Students on the Extent to which School Infrastructures Influence the Affective Ability of Secondary School Students in Enugu State

S/N	Influence of school infrastructures on students' affective ability are to;	Male N=155		Female N=163		Overall		Decision
		$\bar{X}_1$	SD ₁	$\bar{X}_2$	SD ₂	$\bar{X}_G$	SD _G	
12	appreciate lesson taught	1.69	0.61	1.71	0.61	1.69	0.61	LE
13	internalize concepts	1.92	0.83	1.94	0.84	1.93	0.83	LE
14	develop interest in the lessons	1.54	0.64	1.55	0.64	1.54	0.64	LE
15	analyze concepts	1.70	0.72	1.69	0.73	1.70	0.73	LE
16	make description of concepts	1.30	0.46	1.30	0.48	1.30	0.46	LE
17	digest ideas	1.61	0.63	1.61	0.63	1.61	0.63	LE
18	make differentiation of similar issues	1.54	0.64	1.53	0.64	1.54	0.64	LE
19	present questions	1.70	0.61	1.68	0.61	1.69	0.61	LE
20	listen attentively	1.46	0.64	1.47	0.64	1.47	0.64	LE
21	prioritize studies	1.54	0.64	1.55	0.64	1.54	0.65	LE
22	internalize values	1.62	0.63	1.61	0.63	1.62	0.63	LE
23	attaching worth to specific objectives	2.00	0.56	2.01	0.56	2.00	0.56	LE
24	appreciate active participation in class	1.77	0.58	1.77	0.58	1.77	0.58	LE
25	enhanced willingness to listen in class	1.62	0.49	1.62	0.49	1.62	0.49	LE
26	foster deeper learning	1.92	0.73	1.93	0.73	1.93	0.73	LE
Cluster Mean		1.66		1.66		1.66		LE

Note: X=Mean; SD=Standard Deviation; LE = Low Extent; VLE = Very Low Extent

Data presented in Table 3 indicates that the items have mean responses which are lower than the cut-off point in the real limit of numbers depicting low extent. The Standard deviation values are small and closely clustered around the mean. This signifies that the respondents' responses are homogenous. The overall cluster mean rating of 1.66 shows that the respondents rated all the 15 items raised to a low extent. This implies that school infrastructures influence the affective ability of secondary school students in Enugu State to a low.

H0₂. There is no significant difference in the mean ratings of male and female students on the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State.

Table 4: Summary of t-test Analysis of Mean Response Scores of Male and Female Students on the Extent to which School Infrastructures Influence the Affective Ability of Secondary School Students in Enugu State.

VARIABLES	N	t	df	Sig. (2tailed)	Mean Difference	Std. Error Difference	Decision
Male Student	155	.100	316	.92	-.05	.53	Do not reject H0 ₂
Female Student	163						

NS= Not Significant

The result of t-test analysis in Table 4 shows that the t-value at .05 level of significant and 316 degree of freedom for the items is .10 with a significant value of .92. As the significant value of .92 is more than the .05 level of significant the null hypothesis is not significant. This means that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State.

Research Question 3

To what extent do school infrastructures influence the psychomotor ability of secondary school students in Enugu State?

Table 5: Mean Ratings and Standard Deviation of the Male and Female Students on the Extent to which School Infrastructures Influence the Psychomotor Ability of Secondary School Students in Enugu State.

S/N	Influence of school infrastructures on students' psychomotor ability include;	Male N=155		Female N=163		Overall		Decision
		$\bar{X}_1$	SD ₁	$\bar{X}_2$	SD ₂	$\bar{X}_G$	SD _G	
27	making students to be assertive.	3.39	0.49	3.38	0.48	3.38	0.49	HE
28	assisting students in improvising	3.23	0.42	3.23	0.42	3.23	0.42	HE
29	encouraging students to display positive behaviours towards learning	3.31	0.46	3.30	0.46	3.31	0.46	HE
30	helping students in demonstration of concepts	2.85	0.77	2.85	0.78	2.85	0.78	HE
31	assisting students in processing of new ideas	3.00	0.79	2.99	0.78	2.99	0.78	HE
32	helping students to demonstration actions after a visual demonstration	2.99	0.79	3.00	0.79	3.00	0.78	HE
33	helping students to perform action with verbal instruction	3.15	0.77	3.15	0.77	3.15	0.77	HE
34	helping students to perform action with written instruction	3.38	0.63	3.38	0.63	3.38	0.63	HE

35	assisting students in appreciate new ideas	3.23	0.58	3.22	0.58	3.23	0.58	HE
36	assisting students in formulate new ideas	3.00	0.68	2.99	0.68	2.99	0.68	HE
37	influencing students participation in sports	3.39	0.49	3.38	0.49	3.38	0.49	HE
Cluster Mean		3.17		3.19		3.17		HE

Note: X=Mean; SD=Standard Deviation; HE= High Extent

From Table 5 above, the result of data analysis for research question 3 indicated that the respondents rated all the 11 items raised to a high extent with mean scores that are higher than the cut-off point in the real limit of numbers. The variations in the standard deviation of the respondents were insignificant and show unanimity in the responses of the respondents. The value of the grand mean was also high (3.17). This implies that school infrastructures influence the psychomotor ability of secondary school students in Enugu State to a high extent.

H0₃. There is no significant difference in the mean ratings of male and female students on the extent to which school infrastructures influence the psychomotor ability of secondary school students in Enugu State.

Table 6: Summary of t-test Analysis of Mean Response Scores of Male and Female Students on the Extent to which School Infrastructures Influence the Psychomotor Ability of Secondary School Students in Enugu State

Variables	N	t	df	Sig. (2tailed)	Mean Difference	Std. Error Difference	Decision
Male Student	155	.09	316	.92	.03	.36	Do not reject H0 ₃
Female Student	163						

NS= Not Significant

The data obtained from the t-test analysis in Table 6 shows that the t-value at .05 level of significant and 316 degree of freedom for the items is .09 with a significant value of .92. Since the significant value of .92 is more than the .05 level of significant the null hypothesis is not significant. This means that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the psychomotor ability of secondary school students in Enugu State.

Discussion of Findings

Extent to which school infrastructures influence the cognitive ability of secondary school students in Enugu State

The finding revealed that school infrastructures influence the cognitive ability of secondary school students in Enugu State to a high extent. This implies that the respondents agreed to a high extent that school infrastructures influence the cognitive ability of secondary school students in Enugu State. This finding is in consonance with Udoka (2021) who

posited that negative experiences within a students' environment such as school environment has significant impact on the cognitive ability of such student. The finding is in consonance with Jideowolabi (2024) who suggested that infrastructural materials such as school buildings, furniture and fittings, libraries, audiovisuals and laboratories need to be in place, and a regular allocation of resources, including textual, stationeries, chemicals and chalkboards tools are essential for enhancing students' academic performance.

Hence, the learning environment should have a good physical surrounding and physical facilities, which encourage the pupils to concentrate on their studies.

The comparison of the male and female students showed that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the cognitive ability of secondary school students in Enugu State. This is not in alignment with Udoka (2021) who observed that the degree of school students' performance is dependent on gender. He further noted that significant difference exists between male and female students in their performance in school. This is in conformity with Ugwu (2022) who posited that male and female significantly differ in their cognitive ability.

Extent to which school infrastructures influence the affective ability of secondary school students in Enugu State

The finding showed that school infrastructures influence the affective ability of secondary school students in Enugu State to a low. This was supported by the fact that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the affective ability of secondary school students in Enugu State. The finding did not agree with Abugieye (2020) who revealed that school infrastructures enhance students interest will assist them with better ways of handling things emotionally especially as it affects their academics. The findings contradicted the study by Yangambi (2023) who revealed that availability of school physical facilities and

the quality of the school learning environment was said to be powerful factors influencing student success in the emotional and psychomotor domains. The finding did not agree with Edoka (2022) who reported that educational resources which are physical, human, material and symbolic resources influence the academic behaviour of students.

Extent to which school infrastructures influence the psychomotor ability of secondary school students in Enugu State

The finding indicated that school infrastructures influence the psychomotor ability of secondary school students in Enugu State to a high extent. The finding also revealed that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the psychomotor ability of secondary school students in Enugu State. This finding agrees with Chiege (2020) who stated that secondary school students' ability to acquire abilities like psychomotor development has been shown to be strongly impacted by the quality of the physical environment in schools. The finding is in consonance with Nkannem (2021) who noted that without these resources, the classroom environment will not be conducive for effective teaching and learning, which will cause students to become uncomfortable and divert their attention from their studies. The finding however agrees with the study by Iyere (2023) who showed that availability of suitable materials and facilities for psychomotor activities is crucial for the development of motor skills in students.

Conclusion

From the discussion of the findings, it was concluded that school infrastructures influence the cognitive ability, and psychomotor ability of secondary school students in Enugu State to a high extent. The finding revealed that school infrastructures influence the affective ability of secondary school students in Enugu State to a low extent. This is supported by the fact that there is no significant difference in the mean response scores of male and female students on the extent to which school infrastructures influence the cognitive ability, affective ability and psychomotor ability of secondary school students in Enugu State.

Educational Implications of the Findings

The following are the educational implications of this study:

1. The high extent the influence of school infrastructures on the cognitive ability, and psychomotor ability of secondary school students in Enugu State is an indication that there is need to improved infrastructures in secondary schools in Enugu State. Such improvement demand serious efforts of the Government, Ministry of Education and the school authorities so as to improve academic performance of secondary school students in Enugu State.
2. The low extent the school infrastructures influence the affective ability of secondary school students in Enugu State, call for serious concern and increased efforts of the

teachers to enhance the affective skills of the students by making improvisation of instructional aids and learning materials so as to improve academic performance³ of secondary school students in Enugu State.

Recommendations

Based on the findings of the study the following recommendations were made.

1. Government should increase their efforts in provision of infrastructures in secondary school in Enugu State.
2. The secondary school Principals should make more efforts at providing infrastructures within their reach in secondary schools in Enugu State.
3. Deliberate efforts should be made by Ministry of Education in Enugu State at supervising school infrastructures in secondary school in Enugu State.

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